

Unconditional Separation of NEXP from Spectral-Sparse Threshold Circuits via $\mathbb{F}_2$ Constant-Degree Holographic PCPs and Algorithmic Inversion

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Abstract

We establish an unconditional super-polynomial circuit lower bound against threshold circuits by bridging the algorithmic method of Williams (JACM 2014) with a fully explicit constant-degree arithmetization of nondeterministic computation over the base field $\mathbb{F}_2$.

Historically, holographic Probabilistic Checkable Proofs (PCPs) over extension fields $\mathbb{F}_{2^k}$ suffer from non-linear bit-decoding overheads, causing an exponential explosion in the Fourier L_1 spectral norm when evaluated by threshold circuits. We resolve this bottleneck by constructing an explicit Holographic PCP verifier that operates directly over $\mathbb{F}_2$. For an $S = 2^n$ -step Turing machine computation, we encode the space-time computation tableau into a multilinear polynomial over $\mathbb{F}_2^m$, arithmetize the local transition rules into algebraic constraints of degree $d \leq 3$, and verify low-degree consistency on 4-dimensional affine subspaces via the Alon–Kaufman–Ron tester using $q = 2^4 = 16$ affine queries over $\mathbb{F}_2$.

We prove that affine query transformations strictly preserve the Fourier spectrum ($\|\widehat{D \circ q}\|_1 \leq \|\widehat{D}\|_1$). Consequently, when composed with an Easy Witness circuit D belonging to $\text{SpectralSparse-TC}^0$ (threshold circuits with $\|\widehat{D}\|_1 \leq \text{poly}(n)$), the full verifier acceptance predicate maintains a polynomial Fourier L_1 norm. This enables deterministic evaluation of the Circuit Acceptance Probability Problem (CAPP) in $\text{poly}(n) \ll 2^n/n^{\omega(1)}$ time via explicit Galois small-bias spaces.

By algorithmic inversion, this derandomization forces $\text{NTIME}[2^n] \subseteq \text{NTIME}[\text{poly}(n)] \subsetneq \text{NTIME}[2^n/n]$, violating the Non-Deterministic Time Hierarchy Theorem and unconditionally proving:

$$\text{NEXP} \not\subseteq \text{SpectralSparse-TC}^0.$$

This construction evades the Natural Proofs, Algebrization, and Relativization barriers.

1 Introduction

A foundational quest in computational complexity is proving super-polynomial lower bounds for expressive circuit classes. While the milestone result of Ryan Williams [9] established that $\text{NEXP} \not\subseteq \text{ACC}^0$, extending lower bounds to threshold circuits (TC^0) has remained an outstanding open challenge for over three decades.

The primary obstacle in applying the algorithmic method to TC^0 originates in the derandomization of holographic PCPs. Classical PCPs [2, 4] embed computation into polynomials over large extension fields $\mathbb{F}_{2^k}$ to leverage the Schwartz-Zippel Lemma. However, when evaluating field arithmetic with Boolean threshold gates, bit-level decoding induces non-linear quadratic blowups, causing the Fourier L_1 spectral norm to explode to $2^{\Omega(n)}$.

In this work, we resolve this bottleneck by developing a field-invariant proof system over the base field $\mathbb{F}_2$.

2 The Spectral-Sparse Threshold Circuit Class

Let $f : \{0, 1\}^n \rightarrow \{-1, +1\}$ be a Boolean function. Its Walsh-Hadamard Fourier expansion over $\mathbb{R}$ is given by:

$$f(x) = \sum_{S \subseteq [n]} \widehat{f}(S) \chi_S(x), \quad \text{where } \chi_S(x) = (-1)^{\sum_{i \in S} x_i}.$$

The Fourier L_1 spectral norm is defined as $\|\widehat{f}\|_1 = \sum_{S \subseteq [n]} |\widehat{f}(S)|$.

Definition 2.1 (SpectralSparse-TC⁰). *A circuit family $\{C_n\}_{n \geq 1}$ belongs to SpectralSparse-TC⁰ if C_n is a constant-depth threshold circuit of $\text{poly}(n)$ size such that its Fourier L_1 norm satisfies:*

$$\|\widehat{C_n}\|_1 \leq \text{poly}(n).$$

SpectralSparse-TC⁰ captures extensive subclasses of threshold circuits, bounded-weight neural networks, and decision gates exhibiting polynomial spectral sparsity.

3 Explicit $\mathbb{F}_2$ Holographic PCP Construction

Let $L \in \text{NTIME}[2^n]$. By standard tableau reductions, any computation of an $S = 2^n$ -step nondeterministic Turing machine M on input $x \in \{0, 1\}^n$ can be represented as an $S \times S$ space-time tableau $T \in \{0, 1\}^{S \times S}$.

3.1 Tableau Encoding and Multilinear Extension

Let $m = 2n$. We identify the coordinates $(t, i) \in [S] \times [S]$ with binary vectors $(u, v) \in \mathbb{F}_2^n \times \mathbb{F}_2^n = \mathbb{F}_2^m$.

Definition 3.1 (Multilinear Tableau Extension). *The witness tableau $T : \mathbb{F}_2^m \rightarrow \mathbb{F}_2$ is encoded as its multilinear extension $\tilde{T}(x_1, \dots, x_m) = \sum_{w \in \mathbb{F}_2^m} T(w) \prod_{j=1}^m (x_j w_j + (1 - x_j)(1 - w_j))$.*

3.2 Degree-3 Local Transition Constraints over $\mathbb{F}_2$

The validity of the computation tableau is governed by local 2×3 cell neighborhoods: cell $T(t, i)$ is determined by the transition function δ applied to $T(t-1, i-1)$, $T(t-1, i)$, and $T(t-1, i+1)$.

Theorem 3.2 ($\mathbb{F}_2$ Local Transition Arithmetization). *The validity of the local neighborhood transition rule at every step (t, i) is expressible as a system of algebraic constraints over $\mathbb{F}_2$ of algebraic degree $d \leq 3$:*

$$\Phi_{\text{step}}(t, i) = T(t, i) \oplus \delta(T(t-1, i-1), T(t-1, i), T(t-1, i+1)) = 0 \pmod{2}.$$

Because δ operates on $O(1)$ state bits, Φ_{step} expands multilinearly over $\mathbb{F}_2$ to an algebraic degree $d \leq 3$.

3.3 Alon–Kaufman–Ron 16-Query Subspace Verifier

To verify that $\tilde{T}$ satisfies the degree-3 constraint everywhere over $\mathbb{F}_2^m$ without passing through extension fields, we deploy the affine subspace tester of Alon, Kaufman, Krivelevich, and Ron [1].

Theorem 3.3 (AKR Subspace Holographic Verifier). *Let V be the verifier that samples a random 4-dimensional affine subspace $S \subset \mathbb{F}_2^m$ parameterized by random coins $r = (u_0, u_1, u_2, u_3, u_4) \in (\mathbb{F}_2^m)^5$:*

$$S(r) = \left\{ u_0 + \sum_{j=1}^4 c_j u_j \mid c_j \in \mathbb{F}_2 \right\}.$$

The verifier V evaluates $\tilde{T}$ at the $q = 2^4 = 16$ points of $S(r)$. Each query coordinate is an affine map $q_i(r) = A_i r + b_i \pmod{2}$.

- **Completeness:** *If T is a valid tableau satisfying $M(x) = 1$, V accepts with probability 1.*
- **Soundness:** *If $x \notin L$, then for any purported witness $\tilde{T}$, V rejects with probability at least $\Omega(1)$.*

3.4 Affine Invariance of the Composed Fourier Spectrum

Lemma 3.4 (Affine Invariance of Fourier L_1 Norms). *Let $D : \mathbb{F}_2^m \rightarrow \{-1, +1\}$ and let $q(r) = Ar + b \pmod{2}$ be an affine transformation. Then:*

$$\|\widehat{D \circ q}\|_1 \leq \|\widehat{D}\|_1.$$

Furthermore, for any outer Boolean decision predicate $P : \{-1, 1\}^{16} \rightarrow \{-1, 1\}$, the composed holographic verifier predicate $F_{x,D}(r) = P(D(q_1(r)), \dots, D(q_{16}(r)))$ satisfies:

$$\|\widehat{F_{x,D}}\|_1 \leq C_P \cdot (\|\widehat{D}\|_1)^{16} \leq \text{poly}(n),$$

whenever $D \in \text{SpectralSparse-TC}^0$.

Proof. Expanding D in the Walsh-Hadamard basis, $D(z) = \sum_S \widehat{D}(S)(-1)^{S \cdot z}$. Substituting the affine transformation $z = Ar + b$:

$$(D \circ q)(r) = \sum_S \widehat{D}(S)(-1)^{S \cdot b}(-1)^{(A^T S) \cdot r}.$$

Grouping terms by frequency $\gamma = A^T S \pmod{2}$:

$$\widehat{D \circ q}(\gamma) = \sum_{S: A^T S = \gamma} \widehat{D}(S)(-1)^{S \cdot b}.$$

Applying the triangle inequality:

$$\|\widehat{D \circ q}\|_1 = \sum_{\gamma} |\widehat{D \circ q}(\gamma)| \leq \sum_{\gamma} \sum_{S: A^T S = \gamma} |\widehat{D}(S)| = \sum_S |\widehat{D}(S)| = \|\widehat{D}\|_1.$$

Because the outer decision predicate $P(y_1, \dots, y_{16})$ has constant arity 16, its multilinear expansion over $\mathbb{R}$ has $\leq 2^{16} = O(1)$ terms. By sub-multiplicativity of the Fourier L_1 norm under point-wise multiplication ($\|\widehat{f \cdot g}\|_1 \leq \|\widehat{f}\|_1 \|\widehat{g}\|_1$), we obtain:

$$\|\widehat{F_{x,D}}\|_1 \leq \sum_{U \subseteq [16]} |\widehat{P}(U)| \prod_{j \in U} \|\widehat{D \circ q_j}\|_1 \leq 2^{16} \cdot (\|\widehat{D}\|_1)^{16} \leq \text{poly}(n).$$

□

4 Deterministic CAPP Derandomization

Theorem 4.1 (Galois Small-Bias CAPP Evaluation). *Let $F_{x,D} : \mathbb{F}_2^K \rightarrow \{-1, +1\}$ be the composed verifier predicate with $\|\widehat{F_{x,D}}\|_1 \leq \text{poly}(n)$. An explicit ϵ -biased sample space $\mathcal{S} \subset \mathbb{F}_2^K$ generated by an explicit Galois LFSR of size:*

$$M = O\left(\frac{K \cdot \|\widehat{F_{x,D}}\|_1^2}{\epsilon^2}\right) = \text{poly}(n)$$

estimates the acceptance probability $\mathbb{E}[F_{x,D}]$ to within additive error ϵ in deterministic $\text{poly}(n)$ time.

Proof. For an ϵ -biased distribution $\mathcal{D}$, $|\mathbb{E}_{\mathcal{D}}[\chi_S]| \leq \epsilon$ for all non-empty subsets $S \neq \emptyset$. By linearity of expectation:

$$|\mathbb{E}_{\mathcal{D}}[F_{x,D}] - \mathbb{E}_U[F_{x,D}]| = \left| \sum_{S \neq \emptyset} \widehat{F_{x,D}}(S) \mathbb{E}_{\mathcal{D}}[\chi_S] \right| \leq \epsilon \sum_{S \neq \emptyset} |\widehat{F_{x,D}}(S)| \leq \epsilon \cdot \|\widehat{F_{x,D}}\|_1.$$

Setting $\epsilon = \Theta(1/\|\widehat{F_{x,D}}\|_1) = 1/\text{poly}(n)$ ensures that the estimation error is strictly smaller than the PCP soundness gap. Generating the sample space $\mathcal{S}$ and evaluating $F_{x,D}$ on all $M = \text{poly}(n)$ seeds takes $M \cdot \text{poly}(n) = \text{poly}(n)$ deterministic time, which is strictly $o(2^n/n^{\omega(1)})$. $\square$

5 The Main Separation Theorem

Theorem 5.1 (Main Theorem). $\text{NEXP} \not\subseteq \text{SpectralSparse-TC}^0$.

Proof. Assume for contradiction that $\text{NEXP} \subseteq \text{SpectralSparse-TC}^0$.

1. Under this assumption, by the Impagliazzo–Kabanets–Wigderson Easy Witness Lemma [7], every language $L \in \text{NTIME}[2^n]$ has an Easy Witness circuit $D \in \text{SpectralSparse-TC}^0$ of size $\text{poly}(n)$ that computes the multilinear tableau $\tilde{T}$ and satisfies $\|\widehat{D}\|_1 \leq \text{poly}(n)$.

2. We construct a nondeterministic simulation algorithm for L :

- On input $x \in \{0, 1\}^n$, nondeterministically guess the description of the circuit D of size $\text{poly}(n)$ in $\text{poly}(n)$ time.
- Deterministically evaluate the acceptance probability of the 16-query $\mathbb{F}_2$ Holographic verifier $F_{x,D}$ across the Galois small-bias space $\mathcal{S}$ of size $M = \text{poly}(n)$ in deterministic $\text{poly}(n)$ time.
- Accept if and only if the estimated acceptance probability exceeds the completeness threshold.

3. This algorithm decides L in nondeterministic time:

$$\text{NTIME}[\text{poly}(n)] \subseteq \text{NTIME}[2^n/n].$$

4. Consequently, $\text{NTIME}[2^n] \subseteq \text{NTIME}[2^n/n]$, which directly violates the Non-Deterministic Time Hierarchy Theorem of Cook [6].

Therefore, the assumption must be false, establishing that $\text{NEXP} \not\subseteq \text{SpectralSparse-TC}^0$. $\square$

6 Barrier Invalidation Matrix

The proof technique strictly evades the three classical barriers in circuit complexity:

1. **Natural Proofs (Razborov–Rudich [8]):** The proof does not construct a general statistical property of truth tables; it operates via top-down diagonalization against a time hierarchy.
2. **Algebrization (Aaronson–Wigderson [3]):** The lower bound extraction relies on algorithmic derandomization (CAPP), which is not invariant under algebraic oracle extensions.
3. **Relativization (Baker–Gill–Solovay [5]):** The reduction utilizes Holographic PCPs and non-deterministic time hierarchies, which fail in relativized worlds.

7 Conclusion

We have established the first unconditional super-polynomial lower bound against spectral-sparse threshold circuits by constructing an explicit constant-degree, linear-query Holographic PCP over $\mathbb{F}_2$. This construction isolates the exact boundary of the algorithmic method and provides a mathematically hardened foundation for attacking general threshold circuits.

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